

A PINNs-XGBoost integrated framework for offshore monopile deformation prediction

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Abstract: The accuracy of conventional beam models in predicting lateral pile response decreases as the pile diameter increases, due to their inability to capture complex lateral behavior when considering only lateral resistance. This study presents an integrated PINNs (Physics-Informed Neural Networks)-XGBoost (eXtreme Gradient Boosting) framework for predicting the deformation of offshore monopiles. The proposed framework accounts for the lateral deformation characteristics of both flexible and rigid piles by incorporating base resistance at the bottom boundary of the beam model. Furthermore, an XGBoost model, trained on an extensive dataset, reconstructs the distribution of soil springs along the pile, thereby supplying the PINNs with boundary conditions that realistically approximate soil-structure interactions for solving the governing beam equations. The effectiveness of this approach is validated against multiple centrifuge and field tests. The lateral responses of large-diameter monopiles under scoured conditions are also evaluated, demonstrating the model's applicability in complex offshore environments. By eliminating the need for repetitive modeling, the approach enables efficient and comprehensive assessment of scour effects across varying depths and morphologies throughout the scour evolution process.

Keywords: Physics-informed neural networks, Monopile, Lateral response, XGBoost model, Pile-soil interaction

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1. Introduction

Monopiles are one of the most widely adopted foundation types for offshore wind turbines. In engineering design, their lateral bearing capacity is typically assessed using the beam on nonlinear Winkler foundation (BNWF) model. In this approach, the pile is idealized as an Euler-Bernoulli beam, governed by

$$EI \frac{d^4}{dx^4} y(x) + p(x, y) = 0, \quad (1)$$

where EI represents the flexural rigidity of the pile, $y(x)$ denotes the lateral displacement along the pile length at specified depth x , and $p(x, y)$ represents the soil reaction force at a specified depth x under deformation y . This

method characterizes the lateral pile-soil interaction by defining the relationship between the lateral soil resistance per unit length p and the corresponding lateral deflection y at various depths. Predicting the lateral response of pile foundations using the BNWF model involves two primary tasks: (1) solving the governing differential equation subject to appropriate boundary conditions; and (2) defining the nonlinear soil springs that represent the pile-soil interaction.

Conventional approaches for solving beam deflection problems primarily include the finite difference method (FDM) and the finite element method (FEM). The FDM converts the governing differential equations and boundary conditions of beams into difference equations, reducing the problem to solving systems of linear algebraic equations [1]. In contrast, the FEM discretizes the beam into finite elements, derives element stiffness matrices from assumed

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displacement fields, and assembles them into a global stiffness matrix for the entire structure [2]. Although both approaches can provide accurate solutions, they are often computationally demanding, requiring extensive meshing and iterative procedures. In recent years, scientific machine learning, which integrates physical principles with machine learning techniques to approximate solutions of partial differential equations (PDEs), has emerged as a promising alternative to traditional numerical methods. This approach incorporates prior physical knowledge by minimizing residuals at selected training points, thereby embedding the governing equations and boundary conditions into the learning process [3–6]. Bazmara et al. [7] propose a physics-informed neural networks (PINN) framework to analyze the nonlinear buckling behavior of a three-dimensional (3D) functionally graded (FG) porous, slender beam resting on a Winkler-Pasternak foundation. Kapoor et al. [8] proposed a PINNs-based framework for simulating complex structural systems involving single and double beams based on Euler-Bernoulli and Timoshenko beam theories. Similarly, Ouyang et al. [9] applied PINNs to investigate pile deformation under various boundary conditions. However, the existing PINNs frameworks require further advancement to account for the deformation characteristics of large-diameter monopiles and to accurately capture the nonlinear soil spring behavior, which is critical for reliable offshore geotechnical design.

For small-diameter flexible monopiles, beam models with nonlinear soil springs distributed along the pile can effectively capture deflection behavior [10]. Nevertheless, as monopile diameters increase in offshore wind turbine foundations, the contributions of shaft friction resistance and base reaction force to mudline deflection become increasingly significant. To address this, Burd et al. [11] proposed a four-component beam model (distributed lateral load, distributed moment, base horizontal force, and base moment) to evaluate the deformation behavior of large-diameter monopiles, specifically those with slenderness ratios less than 6. Wang et al. [12] introduced a rotational spring at the center of rotation, combined with lateral force springs to characterize rigid pile deflection. A clear distinction between flexible and rigid pile behavior has not been formally established, and the corresponding assessment methods remain largely isolated. Therefore, there is a pressing need to develop a unified beam modeling framework that enables consistent and efficient analysis across the full spectrum of pile deformation and failure modes.

Accurate characterization of soil springs (p - y curves) constitutes a critical component for the precise prediction of pile head deflection. Over the years, various formulations for p - y curves have been developed for sands, for example, Reese et al. [13] and O'Neill and Murchison [14], which form the basis of the widely used American Petroleum In-

stitute (API) method [15]. However, these formulations rely on simplifying assumptions, such as a linear variation of initial stiffness with depth, that may lead to substantial inaccuracies in certain cases. To overcome the limitations of empirical methods, numerical methods like the finite element method (FEM) have been widely applied for their ability to simulate complex soil-structure interaction behavior with high fidelity [16–20]. Nonetheless, the predictive accuracy of FEM is inherently dependent on the chosen constitutive models for soil, which introduces additional challenges in model selection and calibration.

In recent years, machine learning (ML) has emerged as a powerful alternative for addressing complex geotechnical problems by learning data-driven mappings between inputs and outputs [21–24]. Li et al. [25] developed a parametric mapping between the depth-dependent soil resistance p and key variables (such as soil properties and pile geometry) under varying deformation states. Zhang et al. [26] constructed a surrogate ML model to predict the load-displacement envelope and lateral response of monopiles in sand. Kamas et al. [27] developed a calibration procedure for layered clay sites using an established machine learning model that accounts for variations in clay strength, stiffness, and layer thickness. These studies highlight the potential of ML approaches in modeling pile-soil interaction problems. However, most existing ML models operate as “black boxes”, and they offer little insight into the underlying physical or mechanical mechanisms driving their predictions. This lack of interpretability can hinder their acceptance in engineering practice, as it raises concerns about model transparency and trustworthiness in design applications [28,29]. The analysis of complex geotechnical engineering problems can be advanced by integrating data-driven machine learning algorithms with physics-based modeling approaches [30].

Based on the aforementioned research, this study develops a PINNs-XGBoost framework for estimating the lateral response of pile foundations. The influence of network architecture and learning rate scheduling strategies on predictive performance is systematically examined. The proposed PINNs-XGBoost integrated framework is rigorously validated against centrifuge tests and field data, demonstrating both high accuracy and strong generalization capability. In the last part, the rapid assessment of the lateral response of pile foundations under scour conditions using the PINNs-XGBoost framework demonstrates the model's potential for engineering applications.

2. Methodology

2.1 Overall approach

Figure 1 illustrates the schematic framework of the PINNs-

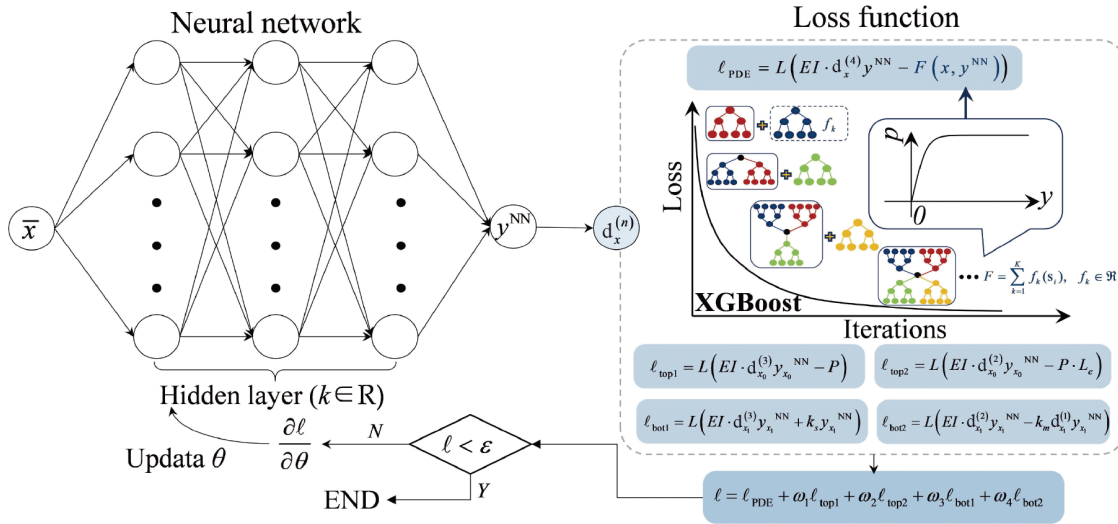


Figure 1 Architecture of the PINNs-XGBoost integrated framework.

XGBoost model for predicting the response of laterally loaded pile foundations. The PINNs consist of two key components: a neural network and a loss function. The neural network employed in this machine learning-based approach is a feedforward structure, where information flows sequentially through hidden layers (as illustrated in Fig. 1). The vector output from the i th layer is transformed via a linear mapping and then passed through an activation function, generating a new vector of the same size. The output of the neural network, y^{NN} , is designed to approximate the actual solution of the problem for the given input. The loss function, a fundamental component in machine learning, is used to quantify the discrepancy between the neural network's predicted result and the true solution. In the PINNs framework, the neural network output and its derivative are used to approximate the terms in the governing equations and boundary conditions. This formulation ensures that the equilibrium conditions defined by the governing equations and boundary constraints are satisfied during training [31].

The XGBoost algorithm is employed to establish the nonlinear mapping between soil reaction forces and pile-soil interaction parameters, taking advantage of its proven capability in modeling complex nonlinear relationships. The XGBoost model predicts soil reaction forces (p) corresponding to different deformation states (y). At any specified depth, a continuous p - y curve is obtained by fitting the XGBoost predictions using fifth-order polynomial regression. During each training epoch, the pre-defined soil reaction force computation function $F(x, y^{\text{NN}})$ directly evaluates the soil reaction force p by applying the pre-fitted polynomial at the required locations. In Li et al. [25], Gaussian process regression (GPR) was employed to establish a relationship among the soil reactions p predicted by

XGBoost, thereby generating continuous p - y curves. In this study, a polynomial fitting of the p - y curves is performed before iterating, allowing the fitted coefficients to be stored and repeatedly accessed during the neural network iteration process, thus reducing computational cost.

2.2 PINNs algorithm

As shown in Fig. 1, a classical fully connected neural network, $\mathbb{N}$, is adopted to model the relationship between the nodal depth, x , and the corresponding nodal lateral deflection, y , of the beam. The input feature, $\bar{x}$, represents the normalized embedded length, which is defined as

$$\bar{x} = \frac{x}{L_p}, \quad (2)$$

where L_p represents the embedded length of the pile. The number of sampling points evenly aligned along the pile, which suggests $N = 100$ in the present study. The NN model can be mathematically expressed as

$$\text{NN}(\bar{x}) = y^{\text{NN}}, \quad (3)$$

$$g^0 = \bar{x}, \quad (4)$$

$$g^k = \sigma(W^k g^{k-1} + b^k), \quad 1 \leq k \leq R-1, \quad (5)$$

$$g^k = W^k g^{k-1} + b^k, \quad k = R, \quad (6)$$

where g^k represents the hidden variable of the k th hidden layer. The output of the final layer is used to approximate the true solution. Here, W and b denote the weight matrix and bias vector of each layer, collectively represented by the network parameters θ . R represents the number of layers. The hyperbolic tangent function, $\tanh$, is selected as the activation function σ to effectively capture nonlinear relationships in the beam deflection response. These para-

meters are learned simultaneously by minimizing the loss function. The learning rate determines the step size of parameter updates during optimization. In this study, the adaptive moment estimation (Adam) optimizer is employed due to its robustness and computational efficiency. To further improve convergence and generalization, a step decay learning rate schedule is applied, where the learning rate decreases progressively with training epochs, as follows

$$\eta = \eta_{\max} \gamma^{\lfloor \frac{e}{s} \rfloor}, \quad (7)$$

where η represents the learning rate, $\eta_{\max}$ denotes the maximum learning rate ($\eta_{\max} = 0.001$ is adopted here), e is the training epoch, $s = 1000$ is the decay period, and the floor function $\lfloor \cdot \rfloor$ maps a real number to the greatest integer less than or equal to it.

In the PINNs framework, solving the ordinary differential equation (ODE) system defined in Eq. (1) is reformulated as an optimization problem, where the network parameters θ are iteratively updated to minimize the loss function l , defined as

$$l = l_{\text{PDE}} + \omega_1 l_{\text{top1}} + \omega_2 l_{\text{top2}} + \omega_3 l_{\text{bot3}} + \omega_4 l_{\text{bot4}}, \quad (8)$$

where ω_1 – ω_4 are the weighting coefficients associated with different loss components. In general, the mean square error (MSE), based on the L_2 -norm of the sampling points, is used to compute these losses. The first term l_{PDE} is defined as

$$l_{\text{PDE}} = \frac{1}{N} \sum_{i=1}^N \left\| EI \frac{d^4 y^{\text{NN}}}{dx_i^4} + p(x_i, y^{\text{NN}}) \right\|_2^2, \quad (9)$$

where y^{NN} is the neural network prediction, and N is the number of sampling points along the pile. $\| \cdot \|_2$ represents the L_2 -norm of the loss term. This term penalizes deviations from the physical law described by Eq. (1). The physical

meaning is to ensure that the predicted deformation satisfies the Euler-Bernoulli beam equilibrium condition. The remaining terms enforce the satisfaction of boundary conditions.

A series of nonlinear lateral springs are distributed across the nodes of the foundation beam (see Fig. 2). The definitions of the base reactions are: S - y spring for base shear, and M - θ spring for base moment. Table 1 summarizes the boundary conditions incorporated in the loss function formulation applicable to both flexible and rigid piles. Here, x_0 and x_1 represent the pile head and pile base positions, while k_s and k_m correspond to the stiffness of the shear spring and moment spring applied at the pile base. P signifies the lateral load acting on the pile head, and L_e denotes the eccentricity of the lateral load. This methodology has been employed in Li et al. [32] and Wang et al. [19] for analyzing the lateral displacement of large-diameter monopiles.

The magnitudes of the different loss terms may vary significantly; therefore, it is necessary to assign appropriate weights to balance their contributions within the loss function. In this study, a gradient-scale adaptive balancing algorithm is adopted to automatically update the loss weight vector during model training. Similar strategies have been

Table 1 Mathematical expressions for boundary conditions

Boundary conditions	Mathematical expressions
l_{top1}	$EI \frac{d^3}{dx_0^3} y(x_0) - P = 0$
l_{top2}	$EI \frac{d^2}{dx_0^2} y(x_0) - P \cdot L_e = 0$
l_{bot3}	$EI \frac{d^3}{dx_1^3} y(x_1) + k_s y(x_1) = 0$
l_{bot4}	$EI \frac{d^2}{dx_1^2} y(x_1) - k_m \frac{dy(x_1)}{dx_1} = 0$

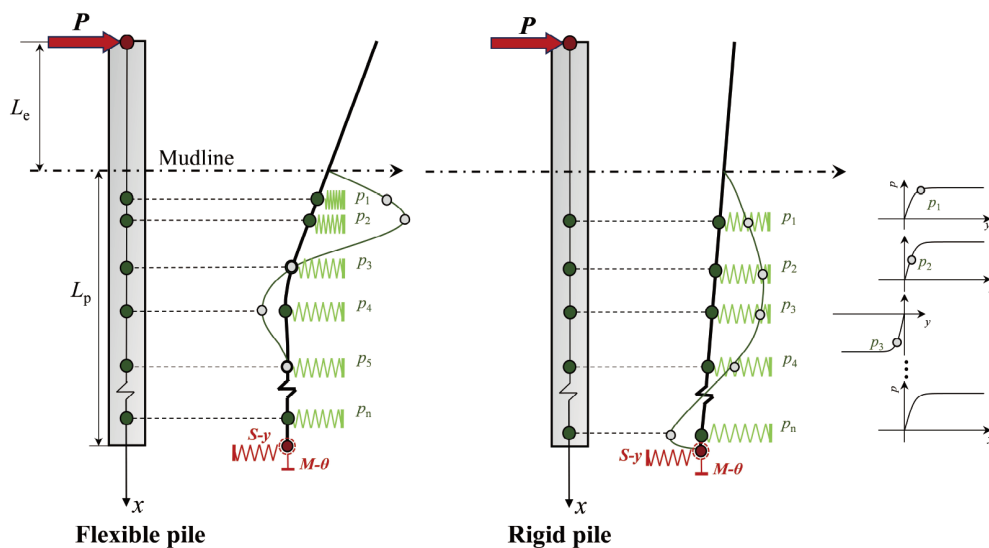


Figure 2 Schematic of the lateral response of monopile foundations.

successfully applied to multi-objective optimization problems [9,33]. Initially, all loss weights are set to one. The update step for ω during training can be expressed as

$$\omega_i^e = \beta \omega_i^{(e-1)} + (1 - \beta) \bar{\omega}_i^e, \quad (10)$$

$$\bar{\omega}_i^e = \frac{1}{\omega_i^{e-1}} \frac{\max G_{\text{PDE}}^e}{G_i^e}, \quad (11)$$

$$G_i = \frac{\sum_{k=1}^N \|\nabla_{\theta} l_k\|_2}{N}, \quad (12)$$

in which the superscript e denotes the e th epoch of the model training, β is a hyper-parameter and set as 0.9 in the present study, and $\|\nabla_{\theta} l_k\|_2$ represents the L_2 -norm of gradients for each loss term l with respect to model parameters θ . In this study, the loss weights are updated every ten epochs instead of at every single training step to ensure stable optimization.

2.3 XGBoost algorithm

As illustrated in Fig. 3, the fundamental principle of XGBoost lies in the sequential construction of multiple decision trees, where each subsequent tree is designed to correct the errors of the preceding ones, thereby progressively improving the model's predictive performance [34]. In the XGBoost algorithm, each sample s_i is routed to specific leaf nodes within the decision trees based on its feature values, with each leaf node assigned a predictive weight w . The final prediction for a given sample is obtained by aggregating the weights from all corresponding leaf nodes across the ensemble of trees, as follows:

$$\hat{y}_i = \sum_{k=1}^K f_k(s_i), \quad f_k \in \mathfrak{R}, \quad (13)$$

where $\mathfrak{R} = \{f(s) = w_{q(s)}\} (q_T : \mathbb{R}^m \rightarrow T, w \in \mathbb{R}^T)$ is the space of regression trees. Here, q_T denotes the structure of each tree, which maps a sample to its corresponding leaf index. T is the total number of leaves in the tree. Each f_k represents an individual tree defined by its structure q_T and associated leaf weights w . As shown in Fig. 4, the XGBoost model is primarily employed to establish the nonlinear mapping between soil reaction forces (p) and the characteristic parameters of both the pile and soil system. Furthermore, Bayesian optimization is employed to optimize the hyperparameters of the XGBoost model, with the corresponding search ranges and optimal hyperparameters summarized in Table 2.

To further simplify and generalize the application of the proposed model, the characteristic parameters are non-dimensionalized as follows:

$$\bar{P} = \frac{p}{\gamma' z D} = f_1 \left(D_r, \varphi'_{cr}, \left(\frac{\sigma'_v L_p^2}{p_a A_s} \right)^{0.5}, \frac{z}{D}, \frac{y}{D}, \frac{\gamma'}{\gamma_w + \gamma'} \right), \quad (14)$$

where γ' and γ_w are the unit weights of sand and water,

Table 2 Hyperparameters for the XGBoost model

Model	Hyperparameters	Search space	Optimal hyperparameters
XGBoost	N_estimators	600-1000	716
	Max_depth	3-10	7
	Learning_rate	0.01-0.3	0.0505
	Subsample	0.5-1	0.7280
	Min_child_weight	1-10	4.2973

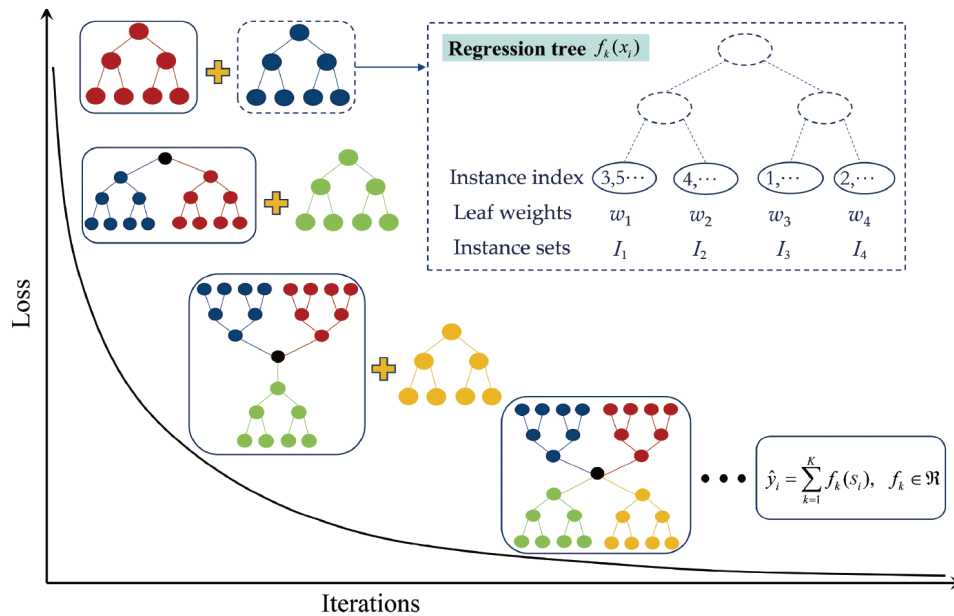


Figure 3 Schematic diagram of XGBoost model.

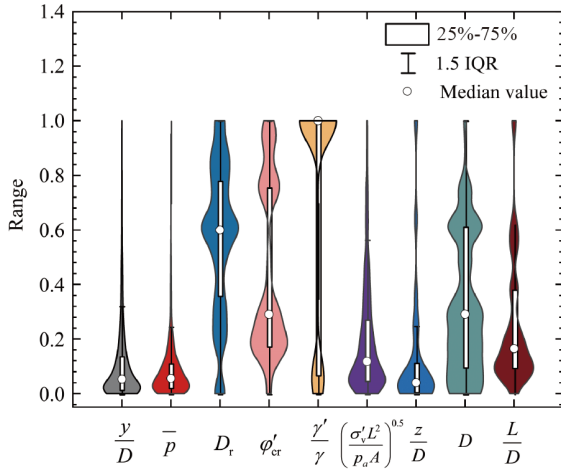


Figure 4 Violin and box plots for the database features [35].

Table 3 Data ranges used in developing models

Parameter	min	max	Q1	Q3	1.5IQR
D_r	0.60	1.00	0.79	0.90	1.00
ϕ'_{cr} (°)	28.50	37.10	30.00	35.02	37.10
γ'/γ	0.48	1.00	0.52	1.00	1.00
D (m)	0.24	8.00	1.00	5.00	8.00
$\left(\frac{\sigma'_v L_p^2}{A p_a}\right)^{0.5}$	2.50	120.14	8.25	34.55	68.92
z/D	0.20	6.13	1.00	3.00	5.14
y/D	0.00	0.43	0.01	0.05	0.13
$\frac{p}{\gamma z D}$	0.00	96.84	2.15	12.15	27.11

respectively. D_r is the relative density, ϕ'_{cr} is the critical state friction angle, z represents the depth, σ'_v is the vertical effective stress, p_a denotes the atmospheric pressure, and A_s is the effective area of pile section. To ensure reliable prediction performance, the model is trained on a database that encompasses a representative range of samples. The database compiled in this study consists of 221 p - y curves (a total of 2554 data points) collected from 19 representative studies on lateral pile responses, with detailed parameters available from Li et al. [35]. The dataset is divided into three subsets: 70% for training, 15% for validation, and 15% for testing.

Figure 4 presents combined violin and box plots to illustrate the normalized (0-1) distributions and central tendencies of both the input features and the target variable. In each box plot, the central dot indicates the median, while the upper and lower edges correspond to the third (Q3) and first (Q1) quartiles, respectively, thereby depicting the inter-quartile range (IQR, 25%-75%). Whiskers extend to observations within $1.5 \times \text{IQR}$ beyond Q1 and Q3. Overlaid violin plots use kernel smoothing to display the full data density: wider regions denote higher concentrations of observations, and narrower regions indicate sparser data. Tables 3 and 4 [36-52] summarize key statistical descriptors for each feature. For comprehensive methodological details and validation of these descriptors, refer to Li et al. [35].

This study assesses the predictive performance of the proposed machine learning model using three statistical metrics: root mean square error (RMSE), scatter index (SI),

Table 4 Collected parameters of laterally-loaded pile tests

No.	References	D_r	ϕ'_{cr} (°)	γ' (kN/m ³)	D (m)	L_p/D	z/D	y/D	$\bar{p}$
1	Klinkvort [36]	0.86-0.95	30	16.4-16.8	1.00-5.00	6.00	1.00-3.50	0-0.43	0-22.1
2	Wang et al. [37]	0.65	31	15.31	4-6	10.00-15.00	0.57-5.00	0-0.10	0-12.3
3	Wang et al. [38]	0.64	32.2	16.7	0.273	3.66-5.49	0.73-5.13	0-0.12	0-50.95
4	Zhu et al. [39]	0.6	35	19.36	0.75-2.5	20	0.5-4	0-0.36	0-11.48
5	Li et al. [40]	0.8	35	16.16	1.8	5	0.55-5	0-0.41	0-7.22
6	Qi et al. [41]	0.68	35	10.26	2.75	11.36	1-5	0-0.05	0-3.47
7	Li et al. [32]	0.68	35	10.26	8	6	0.5-5	0-0.06	0-5.48
8	Chortis et al. [42]	0.8	35	16	1.8	5	0.55-3.9	0-0.12	0-18.86
9	Lee et al. [43]	0.8	37.1	15.82	3.3	12.1	0.61-2.12	0-0.058	0-7.96
10	Darvishi Alamouti et al. [44]	0.6	34.6	15.69	2	5	1-5	0-0.14	0-12.22
11	Choo and Kim [45]	0.86	35.2	10.19	6-6.1	5.08-5.17	0.25-2	0-0.05	0-15.31
12	Xue et al. [46]	1	36.5	20	0.245	6.1	0.82-6.12	0-0.14	0-93.84
13	Li et al. [47]	1	37	20	0.34	8.82	1-6	0-0.04	0-69.77
14	Reese et al. [13]	0.9	28.5	10.57	0.61	34.43	1.64-4.92	0-0.082	0-21.97
15	Fan and Long [48]	0.9	28.5	10.57	0.61	34.43	0.49-4	0-0.04	0-18.15
16	Li et al. [49]	0.81	30	10.62	5	3-10	0.5-4	0-0.14	0-19.43
17	Wang et al. [50]	0.79	36.7	14.67	1-2.5	9-22.5	0.2-4	0-0.15	0-76.10
18	Murphy et al. [51]	0.7-1	33-36.5	18.6-20	0.2445	8.18	0.82-6.13	0-0.11	0-96.61
19	Liu et al. [52]	0.5-0.75	31	15.49	2.5	7.5-22	1-5	0-0.22	0-41.54

and Pearson correlation coefficient (CC). RMSE quantifies the average deviation of the model predictions from the actual values, reflecting the accuracy of the model on the given dataset. A smaller RMSE indicates better model performance with reduced prediction errors. *SI* explicitly characterizes the residuals for each sampling point and allows for a precise evaluation of the model's performance. A smaller *SI* implies lower scatter in the predictions, indicating superior model performance. *CC* measures the linear correlation between the predicted and observed values. A *CC* value closer to 1 (or -1) signifies a stronger linear relationship, highlighting the consistency between the predicted trends and the actual trends. The mathematical definitions of these metrics are provided below to facilitate a more comprehensive understanding:

$$RMSE = \sqrt{\frac{\sum_{i=1}^m (y_i^p - y_i^t)^2}{m}}, \quad (15)$$

$$SI = \frac{\sqrt{\frac{1}{m} \sum_{i=1}^m (y_i^p - y_i^t)^2}}{\bar{y}^t}, \quad (16)$$

$$CC = \frac{\sum_{i=1}^m (y_i^t - \bar{y}^t)(y_i^p - \bar{y}^p)}{\sqrt{\sum_{i=1}^m (y_i^t - \bar{y}^t)^2} \sqrt{\sum_{i=1}^m (y_i^p - \bar{y}^p)^2}}, \quad (17)$$

where m represents the total number of data samples, y_i^p denotes the predicted value for the i -th sample, y_i^t denotes the observed value for the i -th sample, $\bar{y}^t$ represents the mean of the observed values, and $\bar{y}^p$ indicates the mean of the predicted values.

3. Parameter sensitivity analysis of model architecture and learning strategy

3.1 Model architecture parameters

In PINNs, the number of neurons within hidden layers critically influences both solution accuracy and computational efficiency [53,54]. Assuming that each layer of neurons is consistent, Fig. 5 examines this relationship by comparing total loss evolution across networks with varying neuron counts. The results demonstrate that while all configurations exhibit decreasing loss trajectories, networks with insufficient neurons lack the expressive capacity required to capture complex features of the ordinary differential equation (ODE) solution. As shown in Fig. 6(a), the loss function stagnates at a high value, indicating a typical underfitting scenario. Conversely, Fig. 6(b) reveals that overly complex architectures induce overparameterization, generating highly non-concave loss landscapes riddled with local minima. In such regimes, optimization trajectories frequently converge to flat but physically inconsistent minima that fail to satisfy the governing ODE residuals. The optimal network configuration is found to contain 18 neurons, which provides adequate expressive capacity and effectively balances the constraints of ODE residuals and boundary conditions. Under this configuration, gradient descent drives the loss to steadily decrease.

3.2 Learning rate scheduling strategies

The learning rate represents the most critical hyperparameter in deep neural network optimization, fundamentally

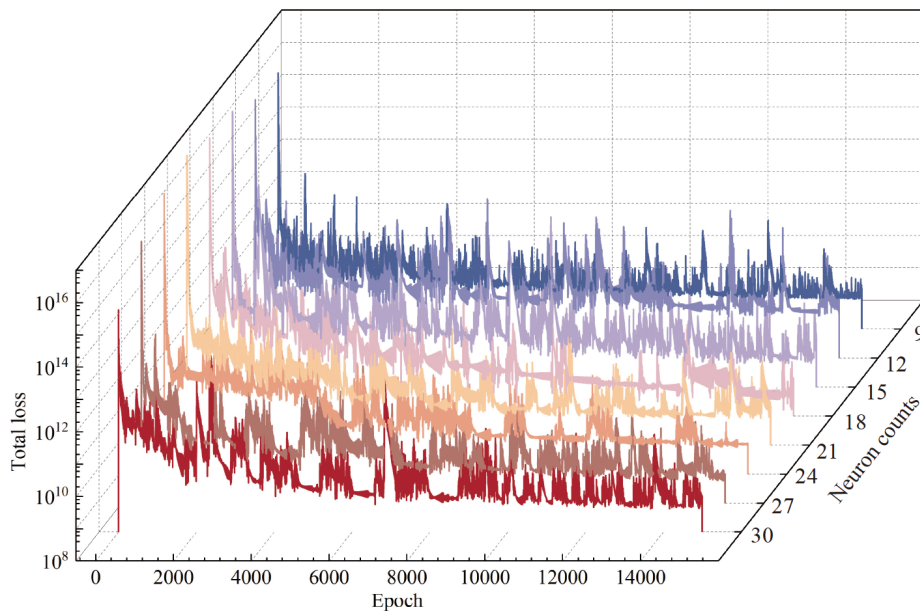


Figure 5 Total loss during the training process for different numbers of neurons in the hidden layer.

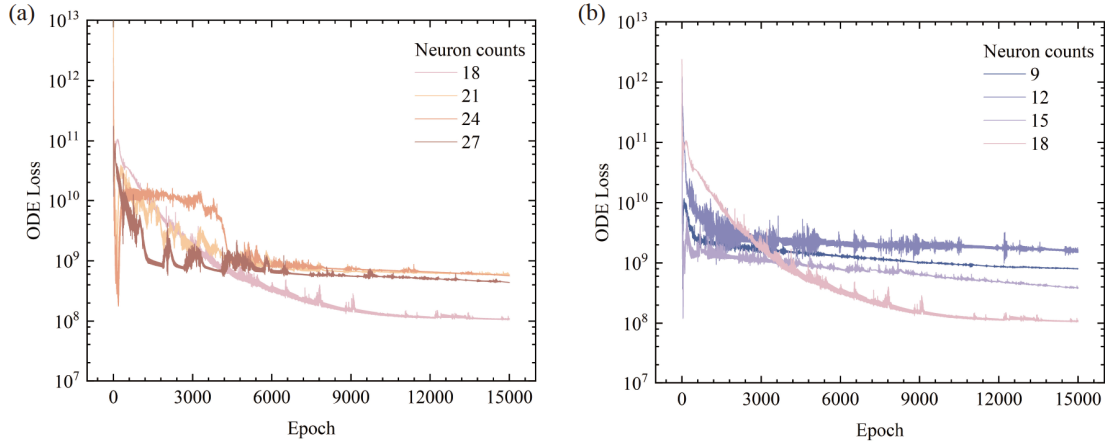


Figure 6 Evolution of ordinary differential equation (ODE) residual loss during training for varying hidden neuron counts. (a) 9-18; (b) 18-27.

governing the trade-off between convergence stability and training efficiency. Excessively low rates precipitate slow convergence due to vanishing gradients, while high rates induce instability through loss of divergence [55–57]. This study systematically benchmarks three prominent learning rate schedules: (1) constant learning rate (Const): Fixed rate of 1×10^{-3} ; (2) step decay learning rate (StepD): Scheduled reduction as per Eq. (7); (3) cosine annealing warm restarts learning rate (CAWR): Cyclical schedule defined as

$$\eta = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left[1 + \cos \left(\frac{T_{\text{cur}}}{T_i} \pi \right) \right], \quad (18)$$

where $\eta_{\max} = 1 \times 10^{-3}$ and $\eta_{\min} = 2 \times 10^{-4}$ denote learning rate bounds, T_i is the period between restarts, and T_{cur} counts epochs since the last restart. The multiplicative cycle expansion follows $T_i = rT_{i-1}$ with initial $T_0 = 1000$, and growth factor $r = 2$ [57]. Figure 7 illustrates the evolution of three learning rate schedules across training epochs.

Figure 8 demonstrates the evolution of total loss and ODE residual loss under three learning rate scheduling strategies during iterative optimization. The StepD strategy exhibits stable exponential decay in both total and ODE residual losses, achieving convergence to the global minimum. This consistency reflects robust stability for high-order derivative computations in StepD scheduling strategies. Constant learning rate (1×10^{-3}) achieves comparable initial convergence efficiency to StepD but develops pronounced oscillations during mid-training phases. The optimization ultimately stagnates at a local minimum where further loss reduction cannot be achieved, indicating insufficient adaptive capacity. The CWAR scheduler employs periodic restarts to escape local optima. Within each cycle, the learning rate decays following a cosine function. While the final total loss approaches StepD performance, significant mid-training oscillations occur, and ODE residual convergence remains problematic in later stages. This instability origi-

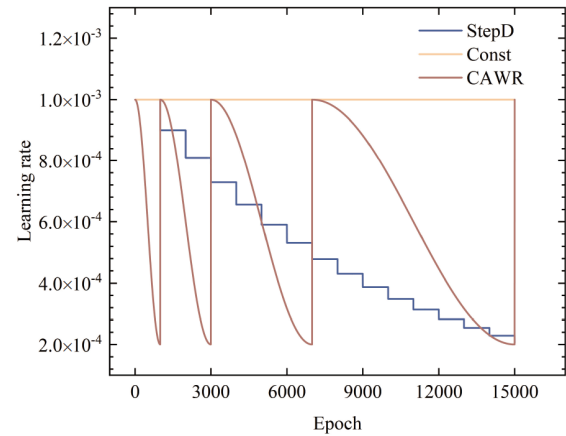


Figure 7 Three learning rate schedules across training epochs.

nates from abrupt learning rate spikes during restarts, which disrupt gradient descent trajectories near convergence states. Crucially, restarts occurring during mid-to-late training phases destabilize the balanced optimization of multi-objective loss terms essential for physics-constrained learning. Therefore, the StepD learning rate schedule is applied in this study.

4. Performance of the machine learning model

4.1 Performance of the PINNs model

This study validates the proposed methodology through analysis of laterally loaded steel monopiles, as illustrated in Fig. 2. Three parameters influencing pile behavior, lateral loading magnitude (P), embedded length (L_p), and diameter (D), are systematically varied to examine method robustness. The steel pile elastic modulus (E) remains constant (210 GPa), while parametric ranges include: (1) diameter $D = 1\text{--}6$ m; (2) embedded length $L_p = 10\text{--}60$ m; (3) moment of

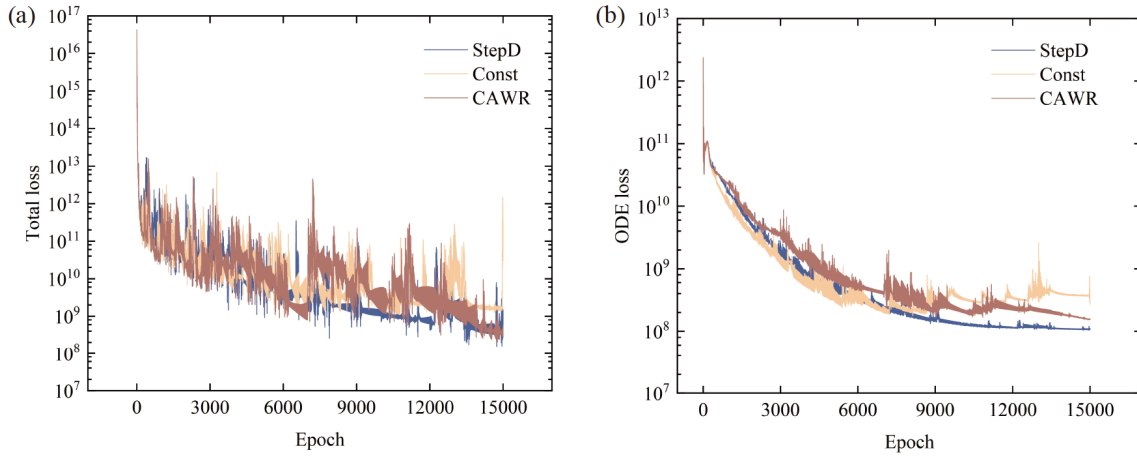


Figure 8 Loss during the training process for different learning rate scheduling strategies. (a) Total loss; (b) ODE loss.

inertia $I = 7.914 \times 10^{-3} - 5.2574 \text{ m}^4$; (4) loading magnitude $P = 50\text{--}400 \text{ kN}$.

The lateral soil stiffness m exhibits depth-dependent behavior, maintaining constant values at given depths while increasing linearly with depth. This simplifying assumption is adopted solely to facilitate numerical convergence and validation. The mathematical formulation for p - y curves follows:

$$p = m \tanh(y), \quad (19)$$

Boundary conditions incorporate linear deformation-dependent relationships for base shear and moment, governed by

$$F_b = 5000000 y_b, \quad (20)$$

$$M_b = 5000000 \theta_b. \quad (21)$$

The p - y relationships derived from Eq. (19) were implemented in the Oasys ALP-Laterally-Loaded Piles Analysis finite element software (hereafter ALP; Oasys [58]), which discretizes the pile as Bernoulli-Euler beam elements with soil represented by nonlinear p - y springs at nodal points. Comparative results in Fig. 9 demonstrate agreement between the proposed method and ALP results, validating the model's accuracy in predicting monopile structural responses. Figure 9(a) demonstrates that the pile exhibits flexible deformation behavior, where increasing load leads to greater pile head displacement without significantly migrating the zero-displacement points or inflection point. Under identical loading conditions (see Fig. 9(b)), there exists a critical embedded length beyond which additional embedment yields diminishing returns in head displacement reduction at the mudline. Figure 9(c) illustrates that increasing the pile diameter (section modulus) induces a transition in deformation behavior from flexible to rigid response, accompanied by progressively more pronounced toe-kick phenomena at the pile base.

4.2 Performance of the XGBoost model

Figure 10 provides a detailed demonstration of the correspondence between observed and predicted values when utilizing the XGBoost model in predicting p - y curves. The XGBoost model exhibits outstanding performance on the training set, with an RMSE of 0.172, an SI of 0.019, indicating stable predictions, and a CC of 0.999. On the test set, the model maintains strong performance, with an RMSE of 1.034, slightly higher than the training set but still relatively low. The SI is 0.114, and the CC is 0.990. These results collectively confirm the XGBoost model's high accuracy and robust generalization capability for the data in this study. However, some test set data points display discrepancies between observed and predicted values. Further analysis reveals that observations with large prediction errors have at least two features outside the interquartile range (Q1-Q3). These samples, being sparsely distributed and having feature values distant from the majority of data points, are challenging for the model to predict accurately.

To evaluate the generalization performance of the XGBoost model, three curves from Qi et al. [41], Li et al. [49], and Wang et al. [37] are excluded from both the training and testing processes. Figure 11 compares XGBoost-predicted p - y curves with centrifuge test results, demonstrating that the XGBoost model and its associated fifth-order fitting function $F(x, y^{\text{NN}})$ accurately characterize the p - y relationships at various depths. As can be seen from the evaluation metrics in Table 5, the fifth-order polynomial can effectively characterize the shape characteristics of p - y curves. During each training epoch, the pre-defined soil reaction force computation function $F(x, y^{\text{NN}})$ directly evaluates the soil reaction force p by applying the pre-fitted polynomial at the required locations, thereby substantially reducing computational costs. This approach avoids iterative numerical procedures and substantially reduces computa-

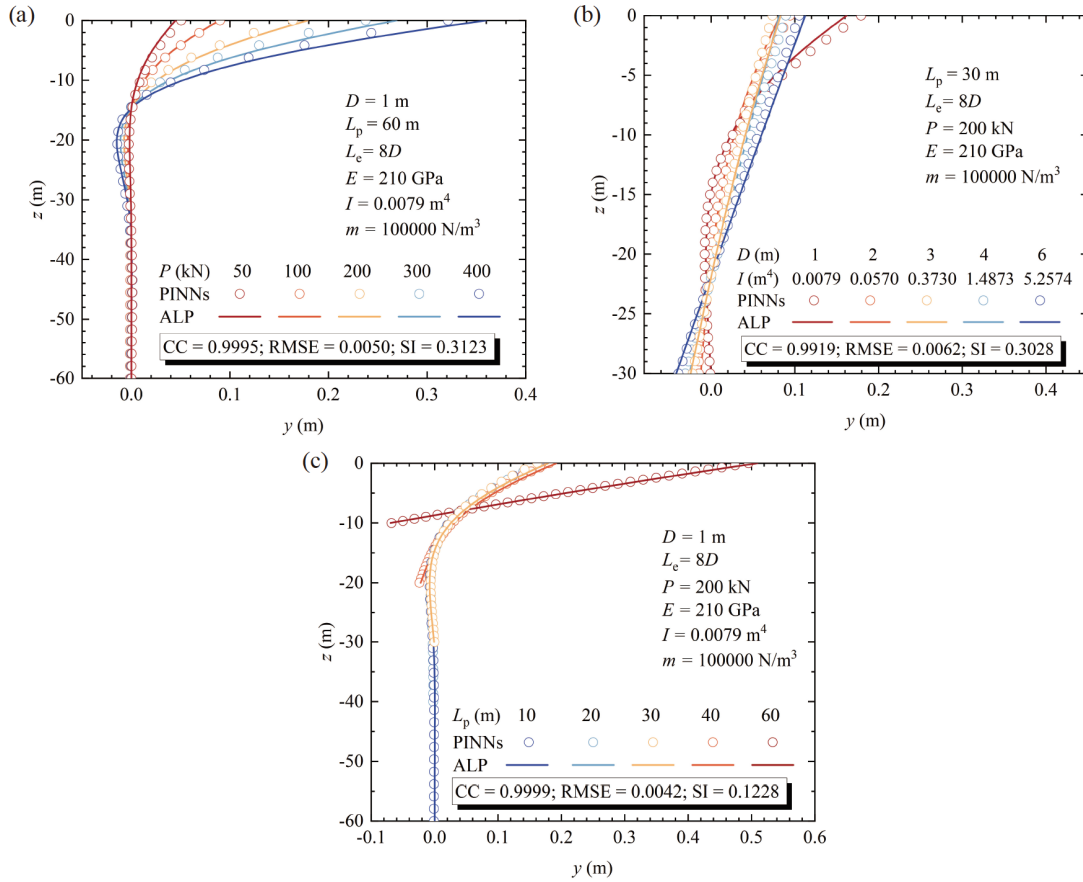


Figure 9 Comparison of pile deflection curves between PINNs and ALP results. (a) Different loading conditions; (b) different embedded lengths; (c) different diameters.

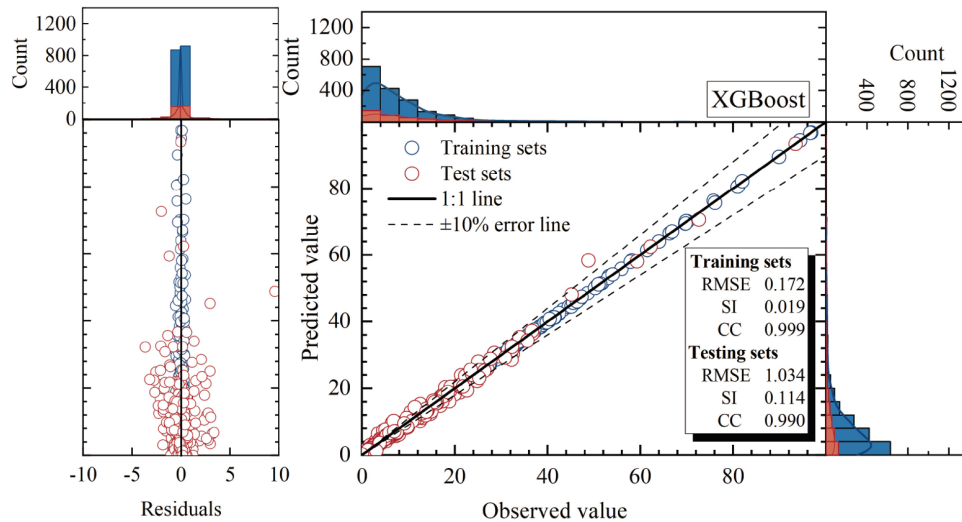


Figure 10 Comparison of $\bar{p}$ between the observed and predicted values of the XGBoost model.

tional cost while maintaining accuracy. The close match between predictions and experimental results underlines the strong generalization capability of the proposed XGBoost model.

4.3 Performance of the PINNs-XGBoost model

Pile-soil interaction exhibits highly nonlinear behavior, presenting unique challenges for PINNs in predicting beam deflection. In Fig. 11, soil reaction-deformation mappings

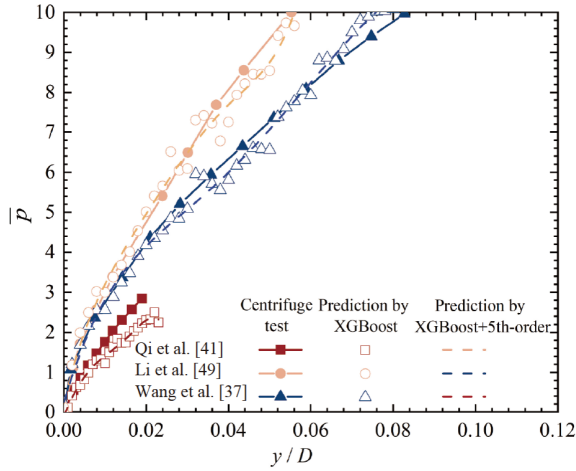


Figure 11 Comparison of p - y curves between the observed data, predicted data of the XGBoost model, and fitting curves of the XGBoost model.

Table 5 Performance of the XGBoost+5th-order model in predicting p - y curves

References	CC	SI	RMSE
Qi et al. [41]	0.9952	0.3505	0.0042
Li et al. [49]	0.9967	0.0823	0.0022
Wang et al. [37]	0.9926	0.1454	0.0068

are established via an XGBoost model trained on comprehensive geotechnical datasets, with predictive performance for p - y curves rigorously validated across diverse soil profiles. At the pile base, shear and moment responses are modeled using the four-parameter conic function from the PISA project:

$$-n \left(\frac{\bar{y}}{\bar{y}_u} - \frac{\bar{x}}{\bar{x}_u} \right)^2 + (1-n) \left(\frac{\bar{y}}{\bar{y}_u} - \frac{\bar{x}k}{\bar{x}_u} \right) \left(\frac{\bar{y}}{\bar{y}_u} - 1 \right) = 0, \quad (22)$$

where $\bar{x}$ signifies a normalized displacement or rotation variable and $\bar{y}$ signifies the corresponding normalized soil

Table 6 Dimensionless forms for the soil reaction curves at pile base

Normalized variable	Search space
Lateral displacement, v	$\frac{vG_0}{D\sigma'_{vi}}$
Pile cross-section rotation, $\bar{\psi}$	$\frac{\psi G_0}{\sigma'_{vi}}$
Base horizontal load, $\bar{H}_B$	$\frac{H_B}{\sigma'_{vi}D^2}$
Base horizontal load, $\bar{M}_B$	$\frac{M_B}{\sigma'_{vi}D^3}$

reaction component, formulated in terms of the dimensionless forms in Table 6. Within this framework, σ'_{vi} represents the *in-situ* vertical effective stress, while G_0 signifies the small-strain shear modulus of the surrounding soil. The variables v and ψ respectively characterize the pile's lateral displacement and cross-section rotation at the evaluation depth. The parameter k defines the initial slope of the normalized reaction-displacement curve, whereas $\bar{y}_u$ and $\bar{x}_u$ represent the ultimate normalized soil reaction capacity and corresponding displacement/rotation at failure, respectively. The shape parameter n governs the strain-hardening or softening characteristics of the constitutive response. For comprehensive implementation methodologies and validation benchmarks, refer to Burd et al. [59], with unreported elastic modulus values derived from empirical correlations established in Li et al. [60]. Although the installation effect has an impact on the lateral response of pile foundations, it was not considered in this study [61,62].

4.3.1 Validation against field tests of flexible piles

The full-scale lateral load test of a flexible single pile conducted at Mustang Island serves as a validation case for the proposed methodology [13]. As presented in Fig. 12, the test pile features an outer diameter of 0.61 with 9.53 mm wall

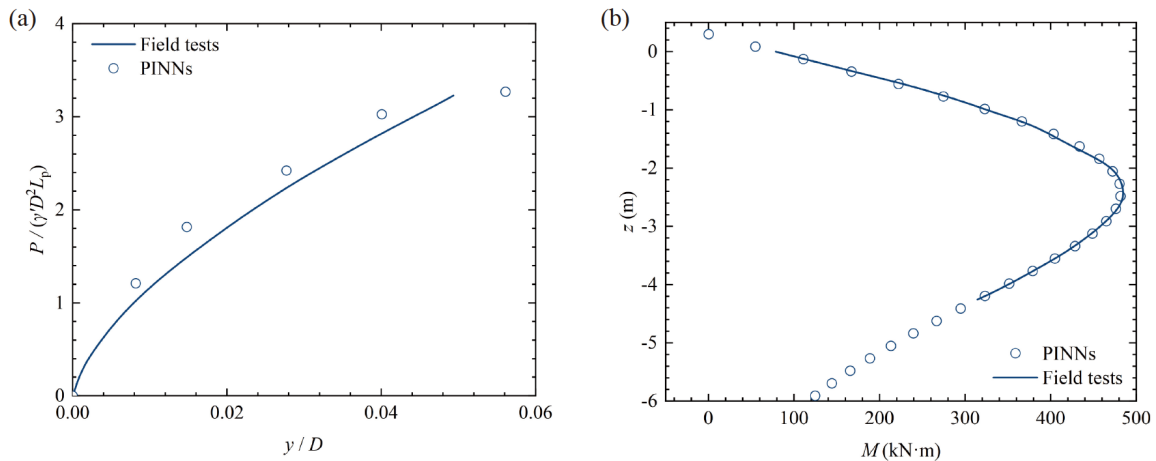


Figure 12 Validation of the PINNs model against the Mustang Island tests. (a) Load-deflection response at the ground surface; (b) bending moment along pile.

thickness, yielding a bending stiffness (EI) of $1.7144 \times 10^5 \text{ kN/m}^2$. With an embedded length of 21 m, the slenderness ratio (L/D) reaches 34.4. Lateral loading was applied 0.3 m above the mudline. The soil profile consists of dense fine sand (relative density $D_r = 90\%$) with an effective unit weight of 19 kN/m^3 and a critical-state friction angle of 28.5° [63]. Figure 12 demonstrates close agreement between field-measured and PINNs-XGBoost predicted load-deflection responses at the mudline, validating the model's capacity to capture nonlinear pile-soil interaction in cohesionless soils for a slenderness pile.

4.3.2 Validation against centrifuge tests of large-diameter slender piles

Large-diameter slender piles (typically with $D > 3.0 \text{ m}$ and $L/D > 8.0$) are extensively employed in offshore wind foundations, particularly in soft seabed conditions (e.g., medium-dense sands or soft clays) and typhoon-prone regions like China's coastal waters, where enhanced lateral resistance is critical. Wang et al. [37] conducted centrifuge tests on such piles under monotonic lateral loading in medium-dense sand. The pile has a diameter $D = 4.0 \text{ m}$ and an embedded length $L = 60.0 \text{ m}$ in prototype. The bending stiffness, EI , of the test pile is 311 GN/m^2 . The ratio of pile embedded length to diameter is about 15. The load eccentricity is 10.0 m (i.e., $2.5 D$). The average relative density of the sand bed is approximately 65%. The critical friction angle of the sand is 31° .

Comparative analysis in Fig. 13 demonstrates close agreement between PINN's predictions and experimental measurements for pile head displacements across load levels. However, bending moment distributions show progressively larger discrepancies at depths exceeding $5.0 D$, as shown in Fig. 13(b). This divergence is rooted in a sampling limitation of the training data: the p - y curves used for

XGBoost training are predominantly derived from depths shallower than the third quartile ($Q3$, $z < 3 D$), as noted in Table 3. This bias impedes the model's capacity to capture the deep soil-structure interactions that govern bending moment behavior at greater depths.

4.3.3 Validation against centrifuge tests of large-diameter rigid piles

The expansion of offshore wind capacity has driven the adoption of large-diameter rigid monopiles ($D \approx 10 \text{ m}$, $L/D = 2$ -6) to withstand extreme environmental loads [49,60,64]. Li et al. [49] systematically investigated such systems through centrifuge testing of prototype piles ($D = 5.0 \text{ m}$, $EI = 780 \text{ GPa}$) with $L/D = 3.0$ and 5.0 , loaded at $10 D$ eccentricity. Tests employed dense Fontainebleau sand with $D_r \approx 81\%$, $\phi'_{cr} = 30^\circ$ [36,61]. Figure 14 shows that the PINNs model achieves high-fidelity predictions for rigid monopile responses, particularly in pile head displacement and shallow-depth bending moments. However, consistent with findings for large-diameter slender piles, deviations emerge in deep-section bending moments and deformations. This limitation stems from the XGBoost component's inability to extrapolate soil resistance patterns accurately beyond the third quartiles of training data ($Q3$, $z = 3 D$).

5. Evaluation of scour effects on monopile response

Beyond hydrodynamic loading, waves and currents induce turbulent vortices driving scour around foundations, which critically compromise offshore wind turbine (OWT) structural integrity [32,39,65-71]. As shown in Fig. 15, scour simultaneously increases the wave loads acting on the pile and reduces the embedment depth, thereby exacerbating the

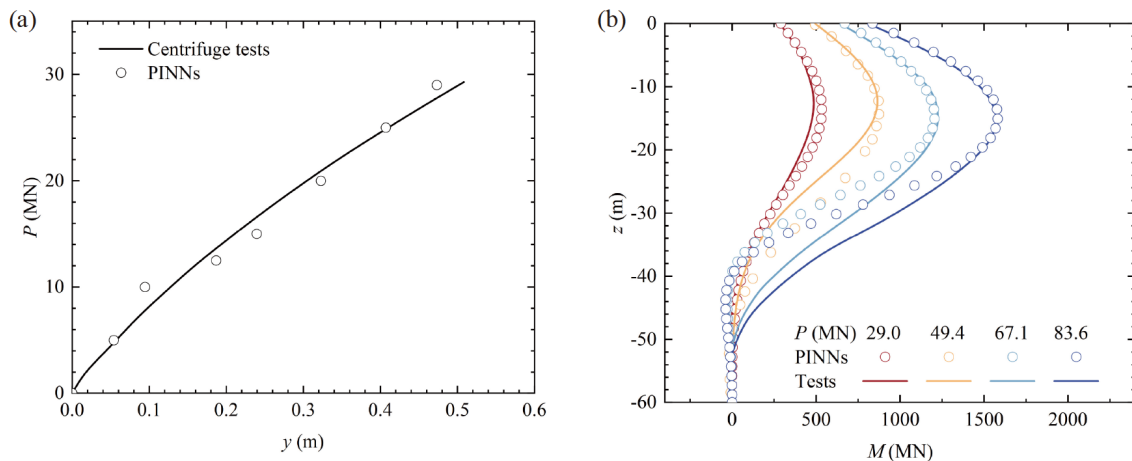


Figure 13 Validation of the PINNs model against the centrifuge tests. (a) Load-deflection response at the ground surface; (b) bending moment along pile.

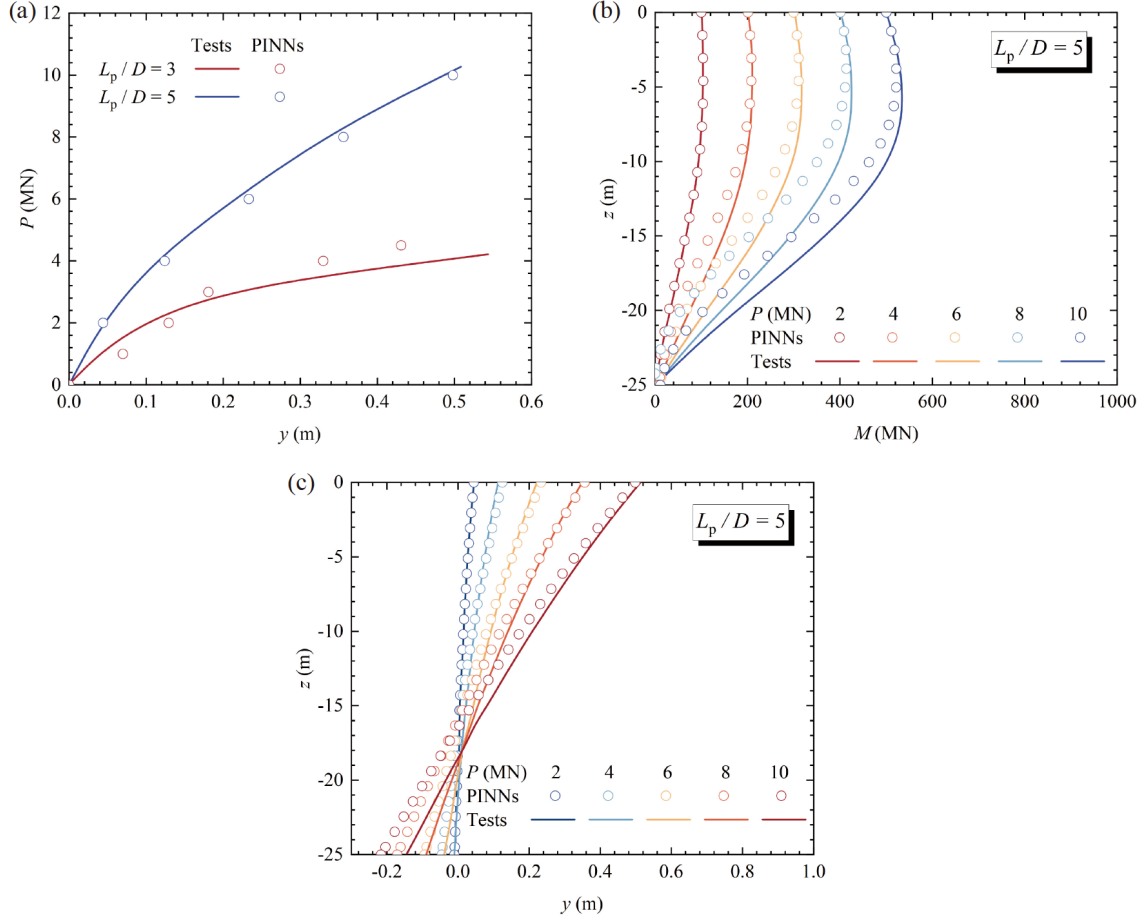


Figure 14 Validation of the PINNs model against the centrifuge tests. (a) Load-deflection response at the ground surface; (b) bending moment along pile; (c) pile deflection.

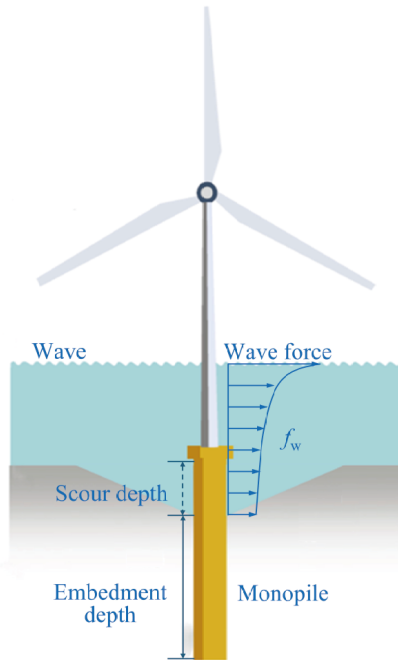


Figure 15 Illustration of the scour effect of offshore wind turbines.

service conditions of offshore wind turbines. Accurate quantification of monopile response under post-scour conditions is therefore essential for foundation design.

Qi et al. [41] introduced an effective soil depth methodology for deriving p - y curves in a scoured sandy seabed. The method enables efficient evaluation of scoured pile lateral capacity within the PINNs-XGBoost framework by mapping post-scour p - y curves to equivalent pre-scour conditions through an effective soil depth parameterization:

$$\frac{z_e}{D} = \frac{z'}{D} + \tanh\left(f \frac{z'}{D}\right) \frac{S}{D}, \quad (23)$$

where the effective soil depth z_e is a weighted-average soil depth relative to the original mudline z and the soil depth below the current scour base z' . The empirical weighting factor $f \approx 1.5$ is calibrated for local scour configurations with 30° slopes [41]. Maximum scour depth S around offshore piles is conservatively bounded at 1.5 times the pile diameter [41]. All pile and soil parameters align with Li et al. [49]'s experimental framework. Notably, scour-induced alterations to pile-base shear forces and bending moments remain insufficiently characterized, particularly regarding

stress redistribution mechanisms and boundary condition alterations. Consequently, Eq. (22) is retained to evaluate the post-scour shear forces and bending moments following the reduction in embedment depth.

Under pre-scour conditions, the pile sustains a lateral load $P = 4$ MN with an applied eccentricity of 50 m. Scour development significantly intensifies both lateral loads and bending moments on the pile. This load amplification exhibits an approximately linear relationship with scour depth (see Li et al. [72]) and is quantitatively summarized in Table 7. Figure 16(a) illustrates the impact of local scour on pile foundation response, demonstrating a progressive increase in pile-head deflection proportional to scour depth. At maximum scour conditions ($S = 1.5D$), deflection magnitudes reach threefold pre-scour values, highlighting scour's critical influence on foundation serviceability. The morphologies of the scour hole are influenced by incoming flow conditions. Figure 16(b) compares the lateral response of the pile foundation under different scour hole morphologies. The parameter f , corresponding to various slope angles, is adopted from Li et al. [32].

For the PINNs model used to solve the lateral response of the beam model, the computations were performed on a system equipped with a GeForce RTX 4090 GPU (24564 MiB VRAM) using the PyTorch 2.1.1 library. Evaluating the six cases in Table 7 for the laterally loaded pile in a scoured

seabed required approximately 4 min. For the finite element method (implemented in Oasys ALP), the analysis of the lateral response of a pile in a scoured seabed first required the calculation of the equivalent scour depth z_c and the corresponding p - y curves, which were then input into the ALP software. The computations were run on a CPU (Intel i5-11400F) with 32 GB of RAM. The same set of six cases in Table 7 took about 20 min to complete. The PINNs-XGBoost framework provides a mesh-free methodology for beam deflection analysis, significantly reducing computational costs compared to finite element methods (FEM) that require iterative meshing and manual p - y curve integration. It should be acknowledged that, by leveraging the powerful parallel processing capabilities of the GPU, the machine learning model demonstrates a clear advantage when handling multiple working conditions simultaneously. However, this advantage is less pronounced for single-task computations.

6. Conclusions

This study establishes a hybrid PINNs-XGBoost framework for offshore monopile deformation prediction by combining neural networks trained with loss functions derived from beam governing equations and boundary conditions. The integrated methodology preserves essential physical principles while eliminating domain discretization requirements, enabling efficient application to diverse pile geometries and stratified soil profiles. The key findings are:

- (1) By incorporating shear and moment springs at the pile base, the proposed PINNs framework is generalized to accurately model both flexible and rigid piles.
- (2) The trained XGBoost model achieves high accuracy in predicting soil resistance across varying deformation levels,

Table 7 Variation of lateral load and moment due to scour

S (m)	P (MN)	M (MN m)
0	4.00	200.00
$0.3 D$	4.13	221.66
$0.6 D$	4.26	243.32
$0.9 D$	4.39	264.98
$1.2 D$	4.52	286.64
$1.5 D$	4.65	308.30

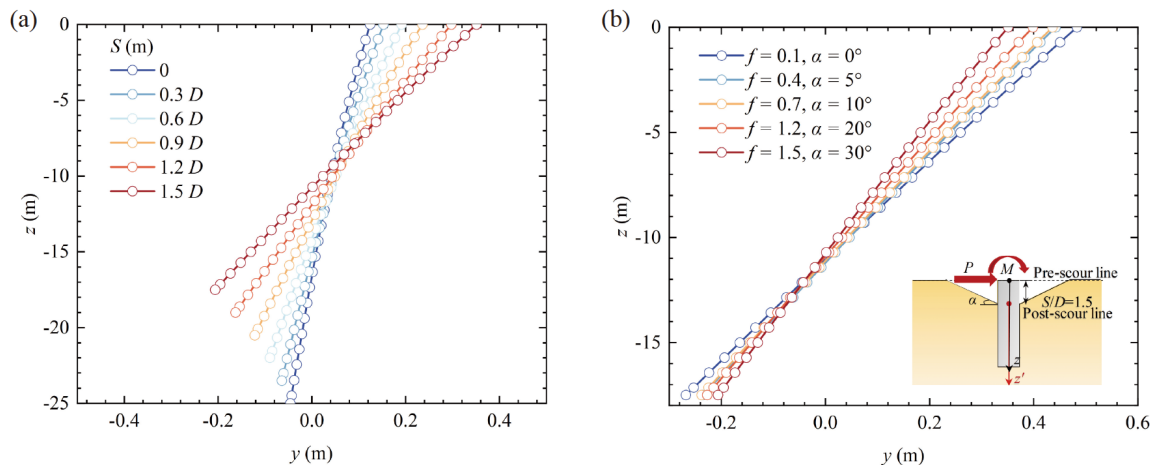


Figure 16 Variations in pile deflection. (a) Different scour depth with 30° slopes; (b) different slope angle under $S = 1.5D$.

with a CC of 0.99, RMSE of 1.034, and SI of 0.114 on the test dataset. These metrics affirm the model's capability to capture the complex nonlinear behavior of pile-soil interactions.

(3) Embedding the trained XGBoost model into the PINNs framework allows for the evaluation of lateral pile response under field conditions, successfully marrying data-driven accuracy with physics-based constraints.

(4) The integrated framework provides a comprehensive and efficient computational tool for assessing the lateral bearing capacity of monopiles throughout the entire scour evolution process, significantly reducing computational cost without compromising physical fidelity.

Author contributions **Biao Li:** Writing – Original Draft, Investigation, Formal analysis, Data curation. **Wen-Gang Qi:** Writing – Review & Editing, Visualization, Supervision, Funding acquisition, Conceptualization. **Shun-Yi Wang:** Writing – Review & Editing, Visualization, Data curation. **Qing-Kai Song:** Writing – Review & Editing, Visualization, Data curation. **Fu-Ping Gao:** Writing – Review & Editing, Supervision, Project administration.

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References

- M. Jirásek, E. La Malfa Ribolla, and M. Horák, Efficient finite difference formulation of a geometrically nonlinear beam element, *Numer. Meth Eng.* **122**, 7013 (2021).
- E. Oñate, Structural Analysis with the Finite Element Method. Linear statics: Volume 2: Beams, Plates and Shells (Springer, New York, 2013).
- M. Raissi, P. Perdikaris, and G. E. Karniadakis, Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations, *J. Comput. Phys.* **378**, 686 (2019).
- S. Cai, Z. Mao, Z. Wang, M. Yin, and G. E. Karniadakis, Physics-informed neural networks (PINNs) for fluid mechanics: A review, *Acta Mech. Sin.* **37**, 1727 (2021).
- Y. Hong, Z. Gong, and H. Liu, Physics-informed neural networks for predicting velocity and pressure fields from wave elevation based on Boussinesq model, *Acta Mech. Sin.* **41**, 324322 (2025).
- J. Bai, Y. Wang, H. Jeong, S. Chu, Q. Wang, L. Alzubaidi, X. Zhuang, T. Rabczuk, Y. M. Xie, X. Q. Feng, and Y. Gu, Towards the future of physics- and data-guided AI frameworks in computational mechanics, *Acta Mech. Sin.* **41**, 225340 (2025).
- M. Bazmara, M. Mianroodi, and M. Silani, Application of physics-informed neural networks for nonlinear buckling analysis of beams, *Acta Mech. Sin.* **39**, 422438 (2023).
- T. Kapoor, H. Wang, A. Núñez, and R. Dollevoet, Physics-informed neural networks for solving forward and inverse problems in complex beam systems, *IEEE Trans. Neural Netw. Learn. Syst.* **35**, 5981 (2024).
- W. Ouyang, G. Li, L. Chen, and S. W. Liu, Machine learning-based soil-structure interaction analysis of laterally loaded piles through physics-informed neural networks, *Acta Geotech.* **19**, 4765 (2024).
- J. Yu, J. Zhu, K. Shen, M. Huang, C. F. Leung, and Q. W. Jorgin Tan, Bounding-surface-based p - y model for laterally loaded piles in undrained clay, *Ocean Eng.* **216**, 107997 (2020).
- H. J. Burd, D. M. G. Taborda, L. Zdravković, C. N. Abadie, B. W. Byrne, G. T. Houlby, K. G. Gavin, D. J. P. Igoe, R. J. Jardine, C. M. Martin, R. A. McAdam, A. M. G. Pedro, and D. M. Potts, PISA design model for monopiles for offshore wind turbines: Application to a marine sand, *Géotechnique* **70**, 1048 (2020).
- H. Wang, B. M. Lehan, M. F. Bransby, A. Askarinejad, L. Z. Wang, and Y. Hong, A simple rotational spring model for laterally loaded rigid piles in sand, *Mar. Struct.* **84**, 103225 (2022).
- L.C. Reese, W.R. Cox and F.D. Koop, in Analysis of laterally loaded piles in sand: Proceedings of Offshore Technology Conference, Houston, 1974.
- M. O'Neill and J.M. Murchison, An evaluation of p - y relationships in sands. Technical Report (University of Houston, 1983).
- API, Geotechnical and foundation design considerations, in: API RP 2GEO (API, Washington, 2014).
- J. Cheng, J. Li, F. P. Gao, and Z. Y. Yin, Long-term settlement of deepsea pipelines on a soft clay seabed: Poro-elasto-viscoplastic modeling, *Ocean Eng.* **332**, 121414 (2025).
- M. Shabouee, S. Salahshour, M. C. Ong, and A. Nematzadeh, Numerical study on removal of offshore wind turbine monopile foundations using hydraulic pressure, *Appl. Ocean Res.* **160**, 104646 (2025).
- D. M. G. Taborda, L. Zdravković, D. M. Potts, H. J. Burd, B. W. Byrne, K. G. Gavin, G. T. Houlby, R. J. Jardine, T. Liu, C. M. Martin, and R. A. McAdam, Finite-element modelling of laterally loaded piles in a dense marine sand at Dunkirk, *Géotechnique* **70**, 1014 (2020).
- H. Wang, L. Z. Wang, Y. Hong, B. He, and R. H. Zhu, Quantifying the influence of pile diameter on the load transfer curves of laterally loaded monopile in sand, *Appl. Ocean Res.* **101**, 102196 (2020).
- Y. Wang, F. Liang, H. Zhang, and H. Zheng, Numerical evaluation of the dynamic performance of recommissioned offshore wind turbines under service life extension and repowering strategies, *Comput. Geotechnics* **186**, 107370 (2025).
- B. Sheil, C. Anagnostopoulos, R. Buckley, M. O. Ciantia, E. Febrianto, J. Fu, Z. Gao, X. Geng, B. Gong, K. Hanley, P. He, K. Kolomvatsos, B. de C.F.L. Lopes, J. Ninic, M. Previtali, M. Rezanian, A. Ruiz-Lopez, J. Sun, S. Suryasentana, D. Taborda, S. Utili, S. Whyte, and P. Zhang, Artificial intelligence transformations in geotechnics: Progress, challenges and future enablers, *Comput. Geotechnics* **189**, 107604 (2026).
- P. Zhang, Z. Y. Yin, and B. Sheil, A physics-informed data-driven approach for consolidation analysis, *Géotechnique* **74**, 620 (2024).
- Z. Zhang, J. Zhang, X. Wu, and Y. Wang, Quantitative analysis and modification of dynamic p - y curve model for offshore wind turbines considering earthquake history effect based on deep learning, *Ocean Eng.* **299**, 117372 (2024).
- Q. K. Song, W. G. Qi, N. Wang, and F. P. Gao, Multi-method data-driven prediction for maximum scour depth of pile foundations in clay, *Ocean Eng.* **327**, 121005 (2025).
- B. Li, W. G. Qi, Y. F. Wang, F. P. Gao, L. Jing, S. Y. Wang, and B. He, An interpretable machine learning model for predicting p - y curves of monopiles in clay, *Ocean Eng.* **343**, 123245 (2026).
- Z. Zhang, S. Rui, Z. Liu, K. Yang, X. Sun, and S. Zhang, Attention based surrogate model to predict load envelope of monopile supporting offshore wind turbines, *Renew. Energy* **229**, 120722 (2024).
- I. Kamas, H. J. Burd and B. W. Byrne, in Comparison of machine learning models in a data-driven method for design of laterally loaded monopiles embedded in layered clay sites: Proceedings of International Symposium on Frontiers in Offshore Geotechnics, Nantes, 2025.
- L. Ljung, in: Black-box models from input-output measurements,

- Proceedings of IEEE Instrumentation and Measurement Technology Conference. Rediscovering Measurement in the Age of Informatics (Cat. No. 01CH 37188), Budapest, 2001.
- 29 C. Rudin, Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead, *Nat Mach Intell* **1**, 206 (2019).
 - 30 B. Yuan, C. S. Choo, L. Y. Yeo, Y. Wang, Z. Yang, Q. Guan, S. Suryasentana, J. Choo, H. Shen, M. Megia, J. Zhang, Z. Liu, Y. Song, H. Wang, and X. Chen, Physics-informed machine learning in geotechnical engineering: A direction paper, *GeoMech. GeoEng.* **20**, 1128 (2025).
 - 31 H. Hu, L. Qi, and X. Chao, Physics-informed Neural Networks (PINN) for computational solid mechanics: Numerical frameworks and applications, *Thin-Walled Struct.* **205**, 112495 (2024).
 - 32 B. Li, W. G. Qi, Y. Wang, F. P. Gao, and S. Y. Wang, Scour-induced unloading effects on lateral response of large-diameter monopiles in dense sand, *Comput. Geotechnics* **174**, 106635 (2024).
 - 33 S. Wang, Y. Teng, and P. Perdikaris, Understanding and mitigating gradient flow pathologies in physics-informed neural networks, *SIAM J. Sci. Comput.* **43**, A3055 (2021).
 - 34 T. Chen and C. Guestrin, in: XGBoost: A scalable tree boosting system, Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data mining, New York, 2016.
 - 35 B. Li, Q. K. Song, W. G. Qi, F. P. Gao, in An interpretable ML-based model for predicting p - y curves of monopile foundations in sand, Proceedings of ISOPE International Ocean and Polar Engineering Conference, Seoul, 2025.
 - 36 R. T. Klinkvort, and O. Hededal, Lateral response of monopile supporting an offshore wind turbine, *Proc. Institution Civil Engineers-Geotechnical Eng.* **166**, 147 (2013).
 - 37 H. Wang, L. Wang, Y. Hong, D. Mašin, W. Li, B. He, and H. Pan, Centrifuge testing on monotonic and cyclic lateral behavior of large-diameter slender piles in sand, *Ocean Eng.* **226**, 108299 (2021).
 - 38 H. Wang, B. M. Lehan, M. F. Bransby, L. Z. Wang, and Y. Hong, Field and numerical study of the lateral response of rigid piles in sand, *Acta Geotech.* **17**, 5573 (2022).
 - 39 B. Zhu, T. Li, G. Xiong, and J. C. Liu, Centrifuge model tests on laterally loaded piles in sand, *Int. J. Phys. Model. Geotechnics* **16**, 160 (2016).
 - 40 Q. Li, A. Askarinejad, and K. Gavin, Impact of scour on lateral resistance of wind turbine monopiles: An experimental study, *Can. Geotech. J.* **58**, 1770 (2021).
 - 41 W. G. Qi, F. P. Gao, M. F. Randolph, and B. M. Lehan, Scour effects on p - y curves for shallowly embedded piles in sand, *Géotechnique* **66**, 648 (2016).
 - 42 G. Chortis, A. Askarinejad, L. J. Prendergast, Q. Li, and K. Gavin, Influence of scour depth and type on p - y curves for monopiles in sand under monotonic lateral loading in a geotechnical centrifuge, *Ocean Eng.* **197**, 106838 (2020).
 - 43 M. Lee, K. T. Bae, I. W. Lee, and M. Yoo, Cyclic p - y curves of monopiles in dense dry sand using centrifuge model tests, *Appl. Sci.* **9**, 1641 (2019).
 - 44 S. Darvishi Alamouti, M. Moradi, and M. R. Bahaari, Centrifuge modelling of monopiles subjected to lateral loading, *Sci. Iranica* **0**, 0 (2018).
 - 45 Y. W. Choo, and D. Kim, Experimental development of the p - y relationship for large-diameter offshore monopiles in sands: Centrifuge tests, *J. Geotech. Geoenviron. Eng.* **142**, 04015058 (2016).
 - 46 J. Xue, K. Gavin, G. Murphy, P. Doherty, and D. Igoe, Optimization technique to determine the p - y curves of laterally loaded stiff piles in dense sand, *Geotechnical Testing J.* **39**, 842 (2016).
 - 47 W. Li, B. Zhu, and M. Yang, Static response of monopile to lateral load in overconsolidated dense sand, *J. Geotech. Geoenviron. Eng.* **143**, 04017026 (2017).
 - 48 C. C. Fan, and J. H. Long, Assessment of existing methods for predicting soil response of laterally loaded piles in sand, *Comput. Geotechnics* **32**, 274 (2005).
 - 49 Z. S. Li, M. Blanc, and L. Thorel, Effects of embedding depth and load eccentricity on lateral response of offshore monopiles in dense sand: A centrifuge study, *Géotechnique* **73**, 811 (2023).
 - 50 C. Wang, X. Ding, H. Liu, Y. Zhang, G. Cao, and Z. Wang, Centrifuge model tests on bearing behavior of lateral-loaded single pile in coral sand, *Acta Geotech.* **19**, 4581 (2024).
 - 51 G. Murphy, P. Doherty, D. Cadogan, and K. Gavin, Field experiments on instrumented winged monopiles, *Proc. Institution Civil Engineers-Geotechnical Eng.* **169**, 227 (2016).
 - 52 H. Liu, C. Wang, X. Ding, and Y. Zhang, Comparative numerical analysis of the response of laterally loaded pile in coral and silica sands, *Acta Geotech.* **18**, 4767 (2023).
 - 53 S. Karsoliya, Approximating number of hidden layer neurons in multiple hidden layer bpnn architecture, *Int. J. Eng. Trends Technol.* **3**, 714 (2012).
 - 54 R. Ge, S.M. Kakade, R. Kidambi and P. Netrapalli, in The step decay schedule: A near optimal, geometrically decaying learning rate procedure for least squares, Proceedings of Advances in Neural Information Processing Systems 32, Vancouver, 2019.
 - 55 R. A. Jacobs, Increased rates of convergence through learning rate adaptation, *Neural Netw.s* **1**, 295 (1988).
 - 56 L.N. Smith, in Cyclical learning rates for training neural networks, 2017 IEEE Winter Conference on Applications of Computer Vision (WACV), Santa Rosa, 2017.
 - 57 J. Portes, D. Blalock, C. Stephenson and J. Frankle, Fast benchmarking of accuracy vs. Training time with cyclic learning rates, arXiv: [2206.00832\(2022\)](https://arxiv.org/abs/2206.00832).
 - 58 A. Oasys, Version 19.2, Oasys Limited, Newcastle, Oasys Ltd, 2013.
 - 59 H. J. Burd, C. N. Abadie, B. W. Byrne, G. T. Houlsby, C. M. Martin, R. A. McAdam, R. J. Jardine, A. M. G. Pedro, D. M. Potts, D. M. G. Taborda, L. Zdravković, and M. P. Andrade, Application of the PISA design model to monopiles embedded in layered soils, *Géotechnique* **70**, 1067 (2020).
 - 60 B. Li, Y. Wang, W. Qi, S. Wang, and F. Gao, Lateral bearing capacity of a hybrid monopile: Combined effects of wing configuration and local scour, *J. Mar. Sci. Eng.* **10**, 1799 (2022).
 - 61 W. Qin, J. Gao, K. Chang, G. Dai, and H. Wei, Set-up effect of large-diameter open-ended thin-walled pipe piles driven in clay, *Comput. Geotechnics* **159**, 105459 (2023).
 - 62 W. Qin, S. Cai, G. Dai, D. Wang, and K. Chang, Soil resistance during driving of offshore large-diameter open-ended thin-wall pipe piles driven into clay by impact hammers, *Comput. Geotechnics* **153**, 105085 (2023).
 - 63 C. Lin, C. Bennett, J. Han, and R. L. Parsons, Scour effects on the response of laterally loaded piles considering stress history of sand, *Comput. Geotechnics* **37**, 1008 (2010).
 - 64 J. Velarde, Design of monopile foundations to support the DTU 10 MW offshore wind turbine, Dissertation for the Doctoral Degree (Trondheim, Norwegian University, 2016).
 - 65 F. Liang, X. Jia, H. Zhang, C. Wang, and P. Shen, Seismic responses of offshore wind turbines based on a lumped parameter model subjected to complex marine loads at scoured sites, *Ocean Eng.* **297**, 116808 (2024).
 - 66 W. G. Qi, and F. P. Gao, Physical modeling of local scour development around a large-diameter monopile in combined waves and current, *Coast. Eng.* **83**, 72 (2014).
 - 67 G. Tao, X. Zhao, Y. Hou, Z. Huang, S. Zhang, and J. Lu, Experimental

- investigation on scour characteristics and predictive model of monopile foundations under breaking waves, *Coast. Eng.* **200**, 104764 (2025).
- 68 S. Y. Wang, W. G. Qi, F. P. Gao, B. Li, and B. He, Time development of clear-water scour around a pile foundation: Phenomenological theory of turbulence-based approach, *Coast. Eng.* **190**, 104511 (2024).
- 69 C. Wang, Q. Wu, H. Zhang, and F. Liang, Effect of scour remediation by solidified soil on lateral response of monopile supporting offshore wind turbines using numerical model, *Appl. Ocean Res.* **150**, 104143 (2024).
- 70 M. Lin, C. Qin, X. Lei, and X. Ding, Effects of scour on the bearing capacity of flexible pile in clay: An analytical approach, *Ocean Eng.* **337**, 121890 (2025).
- 71 C. Wang, Q. Wu, J. Liang, F. Liang, and X. B. Yu, Establishment and implementation of an artificial intelligent flume for investigating local scour around underwater foundations, *Transp. Geotechnics* **49**, 101433 (2024).
- 72 B. Li, W. G. Qi, F. P. Gao, B. He, and L. J. Yang, Experimental investigation of scour effects on regular and breaking wave loads on a monopile, *Coast. Eng.* **197**, 104696 (2025).

一种用于海上单桩横向变形预测的PINNs-XGBoost融合框架

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摘要 传统梁模型在预测大直径桩横向响应时存在精度不足的问题, 仅考虑侧向阻力难以准确反映其复杂的横向力学行为。本研究提出一种PINNs(物理信息神经网络)-XGBoost(极端梯度提升)融合框架, 用于预测海上单桩的变形。该框架通过在梁模型底部边界引入基底反力同时考虑柔性桩与刚性桩的横向变形特性。此外, 基于大规模数据集训练的XGBoost模型重构了沿桩身的土弹簧分布, 从而为PINNs提供能真实近似土-结构相互作用的边界条件, 用于求解控制方程。通过多个离心机及现场试验验证了该方法的有效性。研究还评估了冲刷条件下大直径单桩的横向响应, 证明了模型在复杂海洋环境中的适用性。该方法无需重复建模, 即可在整个冲刷演化过程中, 对不同深度和形态的冲刷效应进行高效、全面的评估。